

Battery Guardian: Physical Consistency Model

1. Scope

The Guardian observes

$$x_k = (T_{\min,k}, T_{\text{avg},k}, T_{\max,k}, SoC_k)$$

from the battery telemetry stream. The nominal VSS/DBC cycle is

$$\Delta t_{\text{nom}} = 100 \text{ ms},$$

with temperature and SoC quantization of 0.5°C and 0.5 percentage points.

The Guardian is deliberately **not** a detailed electro-thermal battery model. It checks a compact set of closed-form consistency constraints. Numeric values below are demonstrator parameters, not qualification limits for a particular battery chemistry.

For temporal checks, the model uses the **actual elapsed receive time**

$$\Delta t_k = t_k^{\text{recv}} - t_{k-1}^{\text{recv}},$$

not the nominal 100-ms period.

2. Configuration

2.1 Timing and warning policy

Parameter	Value	Meaning
nominal sample period	100 ms	expected VSS/DBC cycle
evaluation period	100 ms	Guardian task period
stale timeout τ_{stale}	500 ms	maximum age of the last valid sample
utilization warning threshold u_{warning}	0.8	warning fraction for continuous limits

For a continuous upper bound,

$$u = \frac{\text{observed}}{\text{limit}}, \quad r = \text{observed} - \text{limit}.$$

Classification:

$$u < 0.8 : \text{no detection}, \quad 0.8 \leq u \leq 1 : \text{WARNING}, \quad u > 1 : \text{VIOLATION}.$$

A warning may therefore have $r \leq 0$.

2.2 Temperature envelope

Parameter	Symbol	Value
absolute minimum	$T_{\text{abs,min}}$	-30°C
absolute maximum	$T_{\text{abs,max}}$	70°C
warning margin	M_{warning}	10°C
reference temperature	T_{ref}	20°C
hot-state temperature	T_{hot}	70°C

The operating thresholds are derived:

$$T_{\text{critical}} = T_{\text{abs,max}}, \quad T_{\text{warning}} = T_{\text{abs,max}} - M_{\text{warning}}.$$

Hence:

- $T_{\text{max}} < 60^\circ\text{C}$: no thermal-limit detection
- $60 \leq T_{\text{max}} < 70^\circ\text{C}$: THERMAL_LIMIT / WARNING
- $T_{\text{max}} \geq 70^\circ\text{C}$: THERMAL_LIMIT / CRITICAL

At exactly 70°C , the state is critical but still inside the absolute envelope. PHYSICAL_TEMP_ABSOLUTE_LIMIT starts only above 70°C .

Configuration invariant:

$$T_{\text{abs,min}} \leq T_{\text{ref}} < T_{\text{hot}} \leq T_{\text{abs,max}}.$$

2.3 Temperature-dependent limits

Normalize the thermal state as

$$\theta(T) = \text{clip} \left(\frac{T - T_{\text{ref}}}{T_{\text{hot}} - T_{\text{ref}}}, 0, 1 \right).$$

With the current parameters,

$$S(T) = 12 - 5\theta(T)$$

is the maximum max-min spread and

$$H(T) = 5 - 3\theta(T)$$

the maximum max-average hotspot deviation.

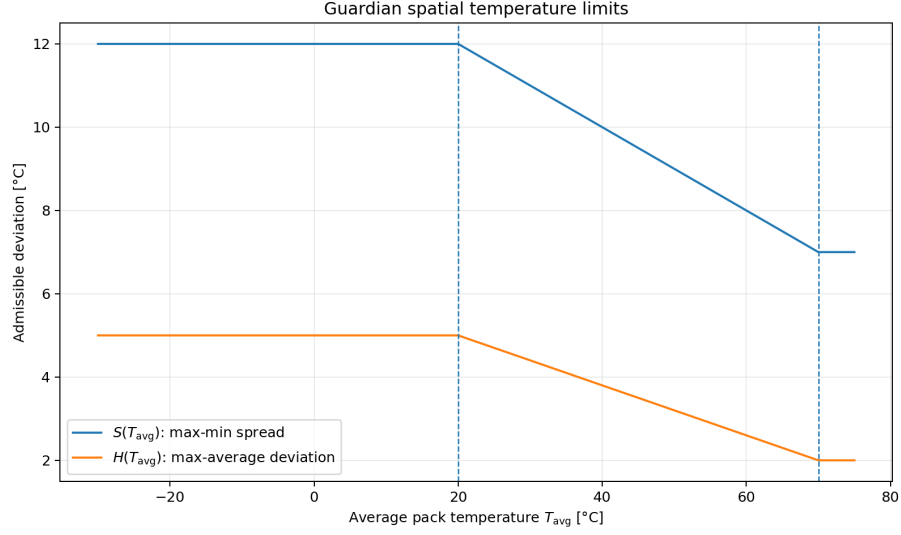


Figure 1: Temperature-dependent spatial limits

Dynamic parameters:

Parameter	Value
heating limit, cold/reference state	8 °C/s
heating limit, hot state	5 °C/s
cooling magnitude limit	6 °C/s
optional SoC coupling gain	0.25 °C/pp
optional SoC-rate cap	2 pp/s

The SoC coupling is **disabled by default**. Gain and cap remain configured so the feature can be enabled without changing the model definition.

2.4 SoC

Parameter	Value
minimum	0 %

Parameter	Value
maximum	100 %
maximum absolute rate	5 pp/s

Using a rate rather than a fixed per-received-sample step avoids false detections after delay or packet loss.

2.5 Stuck detection

Parameter	Value
window length N	10 samples
flatness tolerance ϵ	0.25°C
temperature excitation E_T	1.0°C
SoC excitation E_{SoC}	1.0 pp

Stuck detection applies only to `temp_min`, `temp_avg`, and `temp_max`. SoC is an excitation source, not a stuck target.

3. Detection semantics

A Guardian observation is

`DetectionClass × DetectionLevel`

Canonical classes:

`STREAM_STALE`
`THERMAL_LIMIT`
`PHYSICAL_TEMP_ABSOLUTE_LIMIT`
`PHYSICAL_TEMP_ORDERING`
`PHYSICAL_TEMP_SPREAD`
`PHYSICAL_TEMP_HOTSPOT`
`PHYSICAL_TEMP_RATE`
`PHYSICAL_SOC_RANGE`
`PHYSICAL_SOC_RATE`
`SIGNAL_STUCK`

`WARNING` is used for `THERMAL_LIMIT`, `spread`, `hotspot`, and `temperature rate`. `CRITICAL` is used only for `THERMAL_LIMIT`. All remaining checks are binary and report `VIOLATION`.

4. Static and spatial temperature checks

Absolute envelope

Check independently:

$$T_{\min} \geq T_{\text{abs},\min}, \quad T_{\max} \leq T_{\text{abs},\max}.$$

A strict exceedance raises PHYSICAL_TEMP_ABSOLUTE_LIMIT / VIOLATION.

Ordering

Require

$$T_{\min} \leq T_{\text{avg}} \leq T_{\max}.$$

Violation raises PHYSICAL_TEMP_ORDERING / VIOLATION.

Spread

$$S_{\text{obs}} = T_{\max} - T_{\min}, \quad S_{\text{lim}} = S(T_{\text{avg}}).$$

$$u_S = \frac{S_{\text{obs}}}{S_{\text{lim}}}, \quad r_S = S_{\text{obs}} - S_{\text{lim}}.$$

Class: PHYSICAL_TEMP_SPREAD.

Hotspot

$$H_{\text{obs}} = T_{\max} - T_{\text{avg}}, \quad H_{\text{lim}} = H(T_{\text{avg}}).$$

$$u_H = \frac{H_{\text{obs}}}{H_{\text{lim}}}, \quad r_H = H_{\text{obs}} - H_{\text{lim}}.$$

Class: PHYSICAL_TEMP_HOTSPOT.

Spread and hotspot are separate by design: spread measures pack-wide non-uniformity; hotspot measures divergence of the hottest cell from the thermal bulk.

5. Dynamic temperature consistency

For each $i \in \{\text{min}, \text{avg}, \text{max}\}$,

$$\dot{T}_{i,k} = \frac{T_{i,k} - T_{i,k-1}}{\Delta t_k}.$$

Use the **previous** average temperature for the state-dependent heating limit:

$$R_{\uparrow, \text{base}} = 8 - 3\theta(T_{\text{avg}, k-1}) \quad [^\circ\text{C}/s].$$

If SoC coupling is disabled,

$$R_{\uparrow} = R_{\uparrow, \text{base}}.$$

If enabled,

$$\dot{SoC}_k = \frac{SoC_k - SoC_{k-1}}{\Delta t_k}, \quad Q_k = \min(|\dot{SoC}_k|, 2),$$

$$R_{\uparrow} = R_{\uparrow, \text{base}} + 0.25Q_k.$$

For each temperature channel use

$$(\text{observed}, \text{limit}) = \begin{cases} (\dot{T}_i, R_{\uparrow}), & \dot{T}_i \geq 0, \\ (-\dot{T}_i, 6), & \dot{T}_i < 0. \end{cases}$$

and classify $u_R = \text{observed}/\text{limit}$ with the generic 0.8/1.0 policy.

Class: `PHYSICAL_TEMP_RATE`.

For the nominal 100-ms cycle the base positive step limit is 0.8 °C at the reference state and 0.5 °C at the hot state. This is illustrative only; runtime evaluation uses the actual Δt_k .

The optional coupling is:

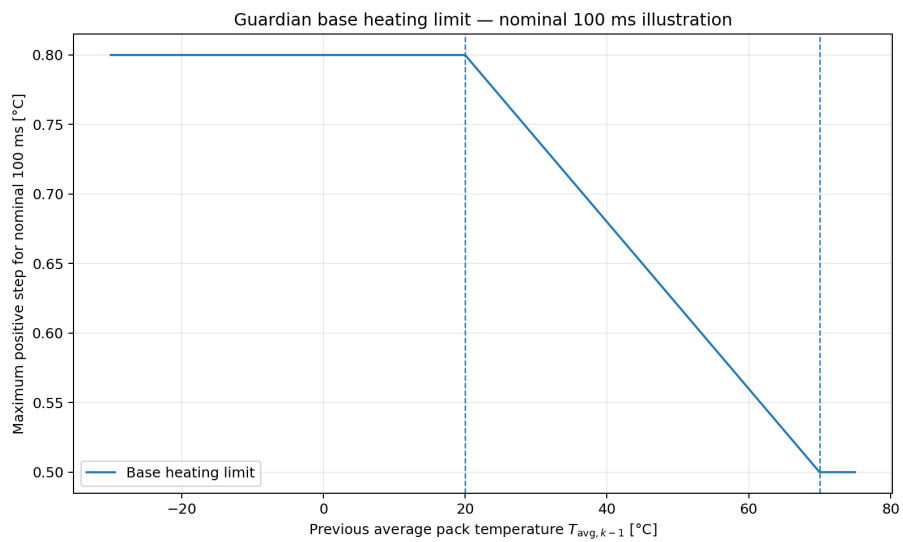


Figure 2: Nominal 100-ms base heating step

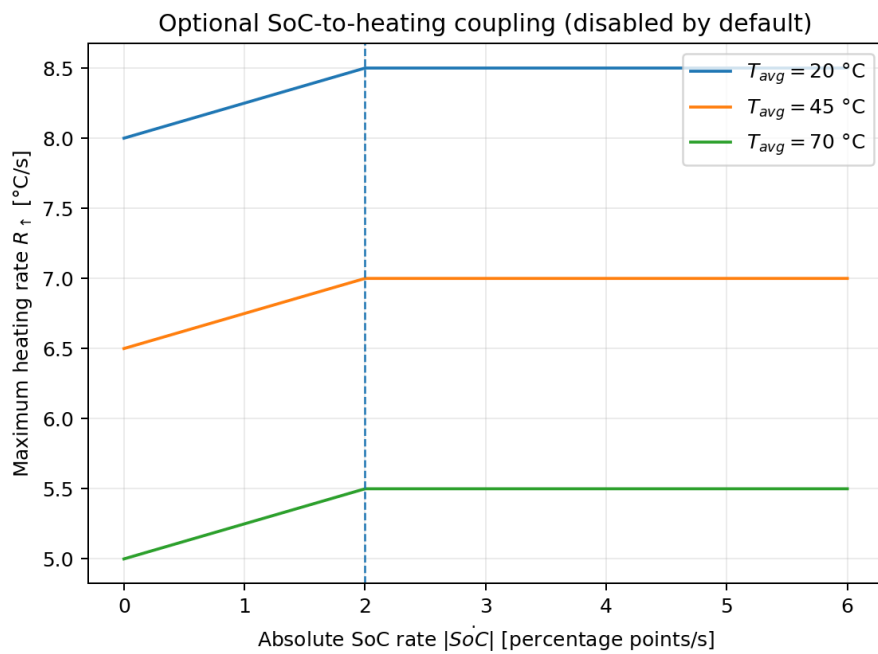


Figure 3: Optional SoC-to-heating coupling

6. SoC consistency

Range

Require

$$0 \leq SoC_k \leq 100.$$

Violation raises PHYSICAL_SOC_RANGE / VIOLATION.

Rate

For consecutive **received** samples,

$$\dot{SoC}_k = \frac{SoC_k - SoC_{k-1}}{\Delta t_k}.$$

Require

$$|\dot{SoC}_k| \leq 5 \text{ pp/s},$$

equivalently

$$|SoC_k - SoC_{k-1}| \leq 5 \Delta t_k.$$

Violation raises PHYSICAL_SOC_RATE / VIOLATION. The first sample has no temporal checks.

7. Stuck detection

For each temperature channel over the last N samples,

$$A_i = \max(T_i) - \min(T_i),$$

and for SoC over the same window,

$$A_{SoC} = \max(SoC) - \min(SoC).$$

A temperature signal is stuck iff

$$A_i \leq \epsilon$$

and there is independent excitation:

$$\max_{j \neq i} A_j \geq E_T \quad \vee \quad A_{SoC} \geq E_{SoC}.$$

A positive result raises **SIGNAL_STUCK / VIOLATION**.

8. Stream freshness and evaluation

Let t_{last} be the receive time of the last valid sample:

$$a(t) = t - t_{\text{last}}.$$

If

$$a(t) > 0.5 \text{ s},$$

raise **STREAM_STALE / VIOLATION**.

Freshness is checked periodically even without new input. Each received sample is physically evaluated at most once; the previous evaluated sample is retained for temporal checks.

The Guardian observes only the symptom:

```
transport.delay  --\
transport.drop   ---+---> STREAM_STALE
source.dropout   --/
```

It does not infer the injected root cause.

9. Evaluation order

For each new valid sample:

1. classify **THERMAL_LIMIT**;
2. check absolute bounds and ordering;
3. evaluate spread and hotspot;
4. if a previous sample exists, evaluate temperature and SoC rates using actual Δt_k ;
5. update and evaluate the stuck window;
6. emit detection state transitions and project configured class/level pairs to DFM.

Malformed messages do not update model state or freshness.

10. Evidence and injection semantics

For continuous checks, evidence should expose at least:

```
detection_class: PHYSICAL_TEMP_SPREAD
detection_level: VIOLATION
observed: 13.0
limit: 9.3
residual: 3.7
utilization: 1.398
```

Temporal evidence should additionally include the affected signal and actual Δt_k .

Injected faults are causes; Guardian detections are observations:

Injection	Typical observation
temperature signal.spike	PHYSICAL_TEMP_RATE
SoC signal.spike	PHYSICAL_SOC_RATE
temperature signal.drift	PHYSICAL_TEMP_SPREAD and/or PHYSICAL_TEMP_HOTSPOT
temperature signal.out_of_range	PHYSICAL_TEMP_ABSOLUTE_LIMIT
SoC signal.out_of_range	PHYSICAL_SOC_RANGE
temperature signal.stuck	SIGNAL_STUCK
transport.delay/drop, source.dropout	STREAM_STALE
signal.combination	zero, one, or multiple detections

The Guardian must never infer the injected class.

11. Compact configuration

```
guardian:
  evaluation_period_ms: 100
  missing_packet_timeout_ms: 500

warning:
  utilization_threshold: 0.8

temperature:
  absolute_min_c: -30.0
  absolute_max_c: 70.0
  warning_margin_c: 10.0
  reference_c: 20.0
  hot_state_c: 70.0
```

```
spread:
  cold_c: 12.0
  hot_c: 7.0

hotspot:
  cold_c: 5.0
  hot_c: 2.0

dynamics:
  heating_rate_c_per_s:
    cold: 8.0
    hot: 5.0
  cooling_rate_c_per_s: 6.0
  soc_coupling:
    enabled: false
    gain_c_per_pp: 0.25
    rate_cap_pp_per_s: 2.0

soc:
  min_percent: 0.0
  max_percent: 100.0
  max_rate_pp_per_s: 5.0

stuck:
  enabled: true
  window_samples: 10
  flatness_epsilon_c: 0.25
  temperature_excitation_c: 1.0
  soc_excitation_pp: 1.0
```